

3 (a)(i)	When two (or more) coherent waves superpose/meet/overlap/exist simultaneously so there is a change in overall intensity (accept: resultant displacement is the sum of the displacement of each wave)	[1] [1]
(a)(ii)	[Any 2] Waves superposing to be coherent or constant phase difference Waves should be of equal amplitude or about the same amplitude Same direction of polarization or unpolarized	[2]
(a)(iii)	Since the 2 waves originate from the same source are in phase and the <u>path difference</u> is a <u>integer multiple of wavelength</u>, or <u>$(n)\lambda$</u>, hence the waves <u>meet in phase</u> at R and <u>constructive interference</u> occurs at R.	[1] [1]
(b)	Minimum path difference = $1 \lambda = 2000 \text{ m}$ (Distance of path taken by reflected wave) – $(2.4 \times 10^4) = 1 \lambda = 2000 \text{ m}$ Distance of path taken by reflected wave = 26000 m	[1] [1]
(c)(i)	$f = \frac{c}{\lambda} = \frac{3 \times 10^8}{2.0 \times 10^3} = 1.5 \times 10^5 \text{ Hz}$	[1]
(ii)	When the incoming frequency matches the natural frequency of the antenna, the antenna will amplify the waves at that radio wave frequency and ignore other waves.	[1]